Abhishek Mehra

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Professional Experience

Umass Chan Medical School — **October 2024 – Present**
Data Engineer — *Remote*

- Spearheaded the deployment of a secure Redshift datawarehouse using **Terraform IaC**, leveraging Redshift's identity-based policies, AWS KMS encryption keys, and VPC security groups to achieve 100% compliance with **HIPAA**
- Designed and deployed a scalable, serverless ETL pipeline on AWS for **a clinical study**, processing **500GB+** of wearable device and participant survey data using S3, AWS Glue (PySpark), and Redshift Serverless
- Orchestrated multi-study workflows with **Apache Airflow (MWAA)** to monitor study-specific S3 buckets, transform data using Python, and load into Amazon Redshift schemas, improving data ingestion efficiency across 5+ studies
- Deployed an automated **ETL framework** to standardize EHR and clinical datasets into OMOP CDM, enabling cross-study analytics; implemented concept-level mapping via OMOP vocabularies (SNOMED) using AWS Glue

Umass Chan Medical School — **July 2023 – December 2023**
Data Engineer Intern — *Boston, MA*

- Led the design of **an event-driven notification system** using Redshift stored procedures, AWS Lambda integration, and SNS, managing critical updates for 50,000+ research participants across 5+ studies
- Developed and productionized a Streamlit-based **BI dashboard** for 5 clinical studies, enabling real-time monitoring of participant engagement metrics and driving data-informed recruitment optimizations
- Led development of a **Python** SDK with unit tests and CI/CD to automate the extraction and transformation of data from REST APIs, optimizing data pipeline workflows, and saving 10 man-hours weekly
- Engineered **SQL Server queries** to integrate clinical and EHR data for patient reporting, delivering real-time analytics for an IBD study in partnership with clinical researchers

Torm Shipping — **July 2018 – September 2021**
Data Engineer — *Mumbai, India*

- Engineered scalable ETL pipelines using **Apache Spark, AWS EMR**, and Redshift for daily processing of vessel telemetry data, analyzing large volumes of ship sensor data (e.g., fuel consumption, engine performance)
- Built a centralized **data lake** on AWS S3 to store and process structured and unstructured maritime data (e.g., weather reports, crew logs), enabling cross-departmental analytics and reducing data retrieval times by 25%
- Created a **Tableau**-based KPI dashboard for fleet performance metrics (e.g., voyage duration, bunker consumption), empowering stakeholders to make data-driven decisions that improved operational efficiency by 12%
- Automated the ingestion of third-party weather API data into Redshift using **AWS Lambda** and Python, enhancing route optimization models that improved fuel efficiency by 6% during adverse weather conditions

Projects

Databricks ETL

- Developed an automated ETL pipeline on **Azure Databricks** to ingest and transform FIFA audience data using **PySpark** and **Delta Lake**, enabling efficient querying and time-travel capabilities for historical data analysis
- Orchestrated data workflows with **Databricks** Jobs and Workflows, incorporating scheduled runs, data validation, and Spark SQL-based aggregations to deliver insights on global audience engagement across FIFA confederations

NLP-based Tweet Classifier

- Developed a machine learning model utilizing **Natural Language Processing** techniques to classify tweets as disaster-related or not, achieving an F1-score of 0.741 using DistilRoberta embeddings
- Designed and deployed an interactive web application using Streamlit, enabling users to input tweets and receive real-time disaster classification results, along with model performance metrics

Education

Northeastern University, *Master's of Science in Analytics* — **July 2024**, *Boston, United States*
Birla Institute of Technology, *Bachelor's in Engineering* — **August 2018**, *Pune, India*

Technical Skills

Languages: Python, SQL, Bash/Shell
Databases: Redshift, Snowflake, PostgreSQL, MySQL, MongoDB, DynamoDB, Star/Snowflake Schema
Data Engineering Big Data Tools: Apache Spark, Kafka, Airflow, dbt, Hadoop, Hive, AWS Glue, EMR, Athena
Cloud Tools: AWS (S3, Redshift, Lambda, RDS, IAM), Terraform, Docker, Git, CI/CD (GitHub Actions, Jenkins)